HP Pavilion a6660t
Restoration, Modernization, and Digital-Preservation Guide

A complete chronological record based only on this conversation

Machine	HP Pavilion a6660t desktop
Final operating system	Lubuntu 24.04 LTS family
Primary storage	WD Blue SA510 2.5-inch SSD, approximately 1 TB
Original storage	Seagate Barracuda 7200.7 ST380013AS, 80 GB
Primary use after restoration	Downstairs shared family computer, legacy-media station, and Rowan Citrix terminal
Record date	August 1, 2026

Prepared for Sanjev Rajaram

No photographs are embedded in this edition.

1. Scope, Source, and Accuracy Standard

This guide is reconstructed exclusively from the current conversation, from its earliest restoration work through the most recent inspection of the removed Seagate hard drive. No other ChatGPT conversation, external document, or web source was used to fill gaps.

The conversation contains both successful work and several incorrect early hypotheses. To preserve the real troubleshooting history without repeating errors as facts, this guide distinguishes among four kinds of information:

Classification	Meaning
Verified	Visible in photographs, BIOS screens, command output, application output, or a successful functional test.
Reported	Stated directly by Sanjev during this conversation and not contradicted later.
Attempted	A command, setting, or troubleshooting path that was actually tried but did not solve the root problem.
Suggested only	An idea proposed in the conversation but not shown as executed. These items are labeled so they are not mistaken for completed work.

Most important correction

The final verified root cause of the recurring boot failure was not a missing operating system and not a permanently required USB boot device. The BIOS repeatedly returned the SATA1 Controller Mode to RAID. This computer boots the installed Lubuntu system correctly when that setting is IDE. Reinstalling GRUB did not solve the failure while the BIOS remained in RAID mode.

Guide contents

2. Executive Summary and Before/After Comparison

The project transformed a 2008-era HP Pavilion a6660t from a slow, unreliable legacy desktop into a usable shared family computer. The completed machine now boots Lubuntu from a modern SSD, connects to current home Wi-Fi through a Wi-Fi 6 USB adapter, supports video calls through a 1080p webcam and microphone, plays audio through external speakers, reads 3.5-inch floppy disks, runs modern web and office applications, and connects to Rowan University through Citrix Workspace.

The restoration was not linear. The system repeatedly produced “Reboot and Select proper Boot device” errors. A Lubuntu installer USB appeared to make the computer boot, which led to a temporary but incorrect theory that the USB had to remain as a permanent boot anchor. Repeated GRUB repairs also appeared successful but did not solve the problem. Full BIOS screenshots finally revealed that the SATA controller was in RAID mode. Changing it to IDE immediately restored normal booting. The same symptom returned whenever the BIOS reverted to RAID, including after the CMOS battery was removed and the BIOS loaded factory defaults.

Area	Before restoration	After restoration
Storage	Original 80 GB Seagate mechanical hard disk; slow by modern standards.	WD Blue SA510 SSD, approximately 1 TB; original HDD removed and preserved externally.
Operating system	Legacy configuration unsuitable for current family use.	Lubuntu 24.04 LTS family, fully updated during the project.
Boot reliability	Recurring boot-device errors and confusing dependence on removable media.	Boots from SSD when BIOS SATA1 Controller Mode is IDE; new CMOS battery installed to retain settings.
Networking	Old Wi-Fi adapter could see networks and connect to a mobile hotspot, but not reliably authenticate to the home network.	TP-Link Archer TX1800U Nano Wi-Fi 6 adapter; old interface disabled; home Wi-Fi works after reboot.
Video calls	No modern webcam setup.	EMEET SmartCam C960 1080p webcam with built-in microphone, verified in Cheese.
Audio	No convenient external speaker setup.	External speakers connected to the green 3.5 mm line-out jack.
Legacy media	No convenient way to inspect family floppies.	External TEAC USB floppy drive; IBM-formatted disks opened; Mac HFS disks identified and investigated.
University access	No Rowan virtual-desktop client.	Citrix Workspace 2604 for Linux installed, authenticated, and opened under the Rowan account.
Family use	Upstairs project machine.	Moved downstairs and configured as a shared family computer with multiple user accounts.

The single most important technical finding

Final root cause

The BIOS factory/default setting was SATA1 Controller Mode = RAID. The installed Lubuntu/GRUB configuration worked when SATA1 Controller Mode = IDE. Whenever the setting returned to RAID, the BIOS could still list the WD Blue SSD on the Main page, yet the machine failed before Linux started and displayed the boot-device error. Returning the controller to IDE restored normal booting.

3. Original Machine and Verified Identifiers

The following specifications were visible in the BIOS, on component labels, in Linux command output, or in later Windows inspection of the removed drive.

Component	Verified identifier or specification	Notes
Desktop	HP Pavilion a6660t	BIOS model number a6660t; product number FK481AV-ABA.
BIOS	Revision 5.33, dated 11/07/08; Core Version 08.00.14	American Megatrends legacy BIOS. F10 enters Setup; Esc opens the boot menu.
Processor	Intel Core 2 Duo E8400 @ 3.00 GHz	BIOS reported 3.00 GHz / 1333 MHz and 6144 KB cache.
Memory	3072 MB PC2-6400 DDR2	Bank 1: 1024 MB; Bank 2: 512 MB; Bank 3: 1024 MB; Bank 4: 512 MB.
Power supply	Lite-On PS-6301-02, 300 W	HP part number 5188-7601.
Optical drive	HL-DT-ST DVD-RAM GH15L	Detected in BIOS as the second drive and CD-ROM boot device.
Monitor	Acer AL1715	Existing external monitor used throughout the project.
Original hard drive	Seagate Barracuda 7200.7, ST380013AS, 80 GB	3.5-inch SATA HDD; firmware 3.43; serial 4MR1RNYF; later removed and examined externally.
Replacement system drive	WD Blue SA510 2.5-inch SSD	Linux reported 931.5 GiB usable capacity, identifying the approximately 1 TB model.
External floppy mechanism	TEAC USB UF000x	Displayed by BIOS in Floppy Group Boot Priority.

Original 80 GB hard drive label details

Field	Value
Series	Seagate Barracuda 7200.7
Model	ST380013AS
Capacity on label	80 Gbytes
Serial number	4MR1RNYF
Part number	9W2812-630
HDA part number	100375322
Configuration level	9WKM1
Configuration code	C6T-01
Firmware	3.43
Date code	05514
HP CPC	5188-1894
HP BPC	371580-002
Interface	SATA

4. Chronological Restoration and Troubleshooting Record

4.1 Project goal and operating-system choice

The machine was not treated as disposable. The goal was to modernize it as far as the hardware reasonably allowed, preserve its useful legacy features, and move it into the downstairs living room as a family computer. Lubuntu was chosen because it is an Ubuntu-based Linux distribution designed to use fewer resources than standard Ubuntu, making it a practical fit for a Core 2 Duo system with 3 GB of DDR2 RAM.

The operating-system family installed was Lubuntu 24.04 LTS. Later system updates brought it to a newer 24.04 point-release state and installed newer kernels, but the base installation remained the Lubuntu 24.04 long-term-support line.

4.2 Creating the Lubuntu installer USB with Rufus

A modern Snapdragon laptop was used to download Lubuntu and prepare a USB installer with Rufus. The installer media was a 64 GB flash drive. Rufus wrote the Lubuntu ISO to the drive so that the HP could boot the installer.

The first USB layout did not appear correctly on the legacy HP. The installer was recreated using a legacy-BIOS-compatible arrangement rather than the more modern GPT/UEFI-oriented layout. In practical terms, the old 2008 BIOS needed an MBR/BIOS-compatible bootable USB. After the USB was rewritten with the compatible scheme, the HP could boot the Lubuntu installer.

Why Rufus mattered

Rufus did more than copy the ISO file. It created a bootable disk structure and allowed the partition scheme and target firmware type to be matched to the older computer. This was necessary because a normal file copy would not make the USB bootable.

4.3 Installing the SSD and Lubuntu

The final system installation was placed on a WD Blue SA510 2.5-inch SSD. Linux later reported the drive as /dev/sda with 931.5 GiB and a single root partition /dev/sda1 mounted at /. The SSD removed the largest performance bottleneck created by the original mechanical disk.

The conversation records that early installation experimentation occurred before the final SSD configuration, and Lubuntu was ultimately installed cleanly to the SSD. The installer used the “erase disk” style of clean installation for the final system. After installation, the machine was updated using the Lubuntu update interface and apt commands.

```
sudo apt update
sudo apt upgrade
```

The SSD greatly improved startup, program launching, updates, web browsing, and general responsiveness. The old processor and 3 GB of RAM still limited heavy multitasking, but storage delays were dramatically reduced.

4.4 Initial Wi-Fi problem: networks visible, home authentication failing

The original wireless adapter could see nearby networks and could connect successfully to a mobile hotspot. It would not reliably connect to the normal home networks. This was a crucial diagnostic result: the wireless hardware was not completely dead, but its combination of chipset, driver, and security support was not compatible with the home router configuration.

The home gateway was an Xfinity XB3 dual-band 802.11ac unit. Changing the router to WPA2-only was discussed, but it was rejected. Sanjev did not want to weaken or broadly change the household network, especially because family members reused short passwords and the same network was used by personal laptops, phones, and a Trezor-connected laptop. The safer choice was to modernize this one computer locally.

Temporary networking alternatives considered

- A long Ethernet cable could provide reliable networking, but the route would cross stairs and living areas and was not practical as a permanent installation.
- A wall telephone jack in the room was examined, but it was an RJ11 telephone connection rather than an Ethernet jack and could not be used directly for networking.
- Changing the router security was rejected, so the mobile hotspot remained a temporary source of Internet access during driver installation and troubleshooting.

4.5 Failed attempts to repair the old Wi-Fi adapter with third-party drivers

The old adapter was associated with the Realtek RTL8812AU family. Multiple driver-source trees were tried. These attempts are important because they explain why purchasing a modern adapter became the rational solution.

Attempt	Observed result	Conclusion
Clone aircrack-ng/rtl8812au from GitHub	git clone failed with an HTTP/2 PROTOCOL_ERROR even though Firefox could browse GitHub.	Browser access and command-line Git transport behaved differently; this was not proof that GitHub itself was unreachable.
Download rtl8812au-5.6.4.2 as a ZIP and install with DKMS	The driver was added to DKMS, but the build failed on kernel 6.17.0-14-generic with “Bad return status” and a Makefile error.	The old source tree was not compatible with the current kernel.
Download morrownr 8812au-20210820-main, driver 5.13.6-23	The DKMS process again failed with build error 10 and reported that the binary package was not found.	A second legacy driver tree also failed on the current system.

No successful custom driver was produced from these attempts. The project stopped trying to force the old chipset into modern compatibility and moved to a newer hardware adapter.

4.6 First major boot-failure cycle and repeated GRUB repairs

The computer began displaying “Reboot and Select proper Boot device or Insert Boot Media in selected Boot device and press a key.” The BIOS Main page could still identify the WD Blue SA510, which made the failure confusing. USB devices, especially the external floppy drive, appeared to change whether the BIOS showed the SSD in HDD Group Boot Priority.

Several GRUB repair methods were attempted. The commands themselves often completed successfully, but they did not solve the boot failure while the BIOS storage controller remained in RAID mode.

Direct live-USB repair attempt

```
sudo mount /dev/sda1 /mnt
sudo grub-install --boot-directory=/mnt/boot /dev/sda
sudo update-grub
```

The direct grub-install command reported success, but update-grub from the live session failed because grub-probe could not obtain the canonical path of /cow. The /cow path belongs to the live “Try Lubuntu” environment, not the installed SSD system.

Chroot repair attempt

```
sudo mount /dev/sda1 /mnt
sudo mount --bind /dev /mnt/dev
sudo mount --bind /proc /mnt/proc
sudo mount --bind /sys /mnt/sys
sudo chroot /mnt
grub-install /dev/sda
update-grub
exit
```

This method entered the installed system and was technically the correct way to run update-grub from a live USB. It still did not fix the recurring boot failure because the BIOS remained configured in the incompatible RAID mode.

Forced GRUB installation

```
grub-install --force /dev/sda
update-grub
```

The forced install also reported “Installation finished. No error reported.” The computer still failed without the USB, confirming that the boot code on the SSD was not the root problem.

4.7 The installer USB appeared to be a boot anchor - but was not the final solution

When the Lubuntu installer USB remained connected, the computer could sometimes boot directly into the Lubuntu installation on the SSD without manually selecting “Try or Install Lubuntu.” The installer USB’s GRUB environment was finding or handing off to the installed system. This led to a temporary theory that a small permanent USB should be left in the computer as a boot helper.

The conversation explored replacing the 64 GB USB with a smaller 4 GB or 8 GB drive and even discussed splitting the 64 GB drive into boot and storage partitions. That workaround was never the correct final repair. Later BIOS evidence proved that the USB only masked the RAID/IDE configuration problem.

Correction to the earlier workaround

A permanent USB was not needed. The correct repair was to set SATA1 Controller Mode to IDE and make sure the BIOS retained that setting.

4.8 External floppy drive and recovery of old files

An external USB floppy drive was attached. BIOS identified it as TEAC USB UF000x. The drive could be plugged into either a front or rear USB port; the front port was convenient for use, while the rear port could be used for a permanent installation.

The first mounting attempt produced “no medium found” because no readable floppy was fully inserted. After the 1.44 MB IBM-formatted disk was inserted in the correct orientation, Lubuntu mounted it at /media/sanjevr/disk and showed its contents.

IBM-formatted floppy results

- A document named story.doc opened successfully in LibreOffice Writer. It contained the childhood story “Death Enterprises Inc.” credited to Sanjev Rajaram under an earlier pen name.
- Other IBM-formatted disks contained school and social-studies projects, personal writing, family documents, and letters relating to medical school.
- The disk also contained JSplit files and temporary files from older Windows-era software.
- Recovered material was copied off the floppies, and a desktop folder named “Floppy Disk Downloads” was used to organize the recovered files.

Mac-formatted floppy investigation

Some Imation disks were labeled “2HD MAC formatted 1.40 MB” rather than “2HD IBM 1.44 MB.” Lubuntu displayed a “Desktop” item or a volume label such as “2MB Mac Format,” which was evidence of an older Macintosh HFS filesystem rather than an empty ordinary folder.

The hfsutils package was installed after the mobile hotspot was turned back on. The USB floppy appeared as /dev/sdf rather than /dev/fd0. After unmounting it from the desktop, hmount recognized the HFS volume.

```
sudo apt install hfsutils -y
lsblk
sudo umount /dev/sdf
sudo hmount /dev/sdf
```

hmount reported volume name “2MB Mac Format,” a creation timestamp of July 17, 1990, and approximately 1,437,184 bytes free. hls did not retain the mounted volume state when run outside the same effective user context, so the HFS inspection remained incomplete. The important conclusion was that the disks were Macintosh-formatted, not necessarily empty or damaged.

4.9 USB devices exposed the boot problem

The external floppy drive and later the webcam appeared to trigger or expose boot failures. When the floppy drive was present, the BIOS sometimes listed the TEAC device in Floppy Group while HDD Group Boot Priority temporarily displayed “Not Installed.” Removing the device did not permanently solve the problem. The USB devices altered BIOS enumeration and made the underlying configuration failure more visible, but the root cause was still the SATA controller mode.

4.10 BIOS deep inspection and the RAID-to-IDE breakthrough

The turning point came when every BIOS page was photographed rather than only the boot-order page. The Advanced page showed:

```
SATA1 Controller Mode = RAID
```

The computer contained one primary SSD and was not using an actual RAID array. Changing the setting to IDE caused additional controller lines to appear and allowed the old BIOS to present the SATA SSD in the legacy-compatible mode expected by this installation.

```
SATA1 Controller Mode = IDE
```

After saving with F10, the computer booted directly into Lubuntu. Multiple forms of reboot were then tested: sudo reboot, the Lubuntu graphical Restart command, a complete shutdown followed by power-on, and a power-loss simulation. All succeeded while the BIOS remained in IDE mode.

Why RAID and IDE mattered on this computer

RAID mode enables a RAID-capable storage-controller presentation, even if only one disk is attached. IDE mode in this BIOS does not change the physical SATA connector into an old ribbon-cable IDE drive; it makes the SATA controller expose a simpler legacy-compatible interface. On a newer machine, AHCI would normally be preferred, but this BIOS and this installed system behaved reliably in IDE mode.

4.11 Installing the modern Wi-Fi adapter

A TP-Link Archer TX1800U Nano AX1800 Wi-Fi 6 USB adapter was purchased from Best Buy. Its purpose was to replace the old adapter's security and protocol limitations without weakening the home router.

Lubuntu initially showed two Wi-Fi interfaces:

```
wlx001644187404
wlxccbabdde9a6a
```

By unplugging and reconnecting the new adapter, the project determined that wlxccbabdde9a6a belonged to the new TP-Link adapter. The old adapter was wlx001644187404. The old device was set to unmanaged so NetworkManager would not use it.

```
sudo nmcli device set wlx001644187404 managed no
```

After rebooting, the new adapter connected successfully to the home Wi-Fi. No manual source-driver installation was ultimately required for the new adapter. The final useful arrangement was: old adapter unmanaged; TP-Link adapter managed and active.

4.12 Webcam and audio

An EMEET SmartCam C960 USB webcam was mounted on the monitor. The box identified it as Full HD 1080p and fixed focus. Lubuntu recognized it through the standard USB video class. Cheese was installed and used to verify live video. The webcam's blue indicator illuminated and the built-in microphone was available.

```
sudo apt install cheese -y
cheese
```

The webcam does not contain output speakers. External speakers were added separately and connected to the computer's green 3.5 mm line-out jack. This completed video-call audio input and output: webcam microphone for input, external speakers for output.

4.13 System updates and conversion into a family computer

Lubuntu offered normal security, kernel, and package updates. The updates were installed and followed by restarts. The computer was moved from the upstairs room to the downstairs living room and repurposed as a shared family computer.

Multiple user accounts were created. The login screen later displayed Guest and named family accounts. The sudo group was checked with getent, and the verified output showed that only sanjevr had administrator privileges:

```
getent group sudo
# Verified output:
sudo:x:27:sanjevr
```

Commands discussed for family account administration included listing normal users, changing another user's password, forcing a password change, and removing a user from the sudo group.

4.14 Thunderbird installation problem

Thunderbird was requested as a Gmail-capable desktop email client. On Ubuntu/Lubuntu 24.04, the apt package named thunderbird was a transitional package that installed the Snap version rather than a native deb package.

```
sudo apt install thunderbird -y
```

The installed transitional package reported version 2:1snap1-0ubuntu3. Thunderbird did not launch from Discover and the shell command thunderbird was initially not found. The Snap was removed, which created small automatic Snap data snapshots. snap saved displayed two Thunderbird snapshots of 153 KB each.

```
sudo snap remove thunderbird
snap saved
```

The Mozilla Team PPA was added, but installing thunderbird still selected Ubuntu's Snap wrapper because package pinning had not yet been successfully completed. The conversation proposed an apt preference file to prefer the PPA, but there was no confirmed successful non-Snap Thunderbird installation. Thunderbird therefore remains unresolved/pending in this record.

4.15 Installing Rowan University Citrix Workspace

The Rowan Information Resources and Technology page identified Citrix as the platform for virtual desktops and applications. The Linux download page offered Citrix Workspace app 2604. The correct package was the x86_64 Debian full package with self-service support, file name icaclient-gcc-8_26.04.0.105_amd64.deb, approximately 324 MB.

Opening the .deb through QApt took a long time and ended with the generic message "Cannot satisfy dependencies." The terminal installation with apt was more informative and successfully installed the missing dependencies and the package.

```
cd ~/Downloads
sudo apt install ./icaclient*.deb
```

During configuration, the installer asked about App Protection, deviceTRUST, and EPA. "No" was selected for all three optional enterprise components. The installation ended with an _apt unsandboxed-access warning because the local .deb was in the user's Downloads folder; this was a warning, not a failure.

The self-service executable was not on the normal shell PATH, but launching it directly worked:

```
/opt/Citrix/ICAClient/selfservice
```

The EULA was accepted, Rowan authentication completed, and Citrix displayed "Good evening, Rajaram, Sanjev." The Home, Apps, and Desktops interface opened. The resource cards loaded very slowly and sometimes remained as placeholders for more than ten minutes.

Performance measurements during Citrix loading

Measurement	Observed value	Interpretation
Installed RAM	2.9 GiB usable	Matches the 3 GB physical configuration.
Memory in use	About 1.4 GiB	The system was not completely out of RAM.
Available memory	About 1.5 GiB	There was still reclaimable/available memory.
Swap	512 MiB total, almost fully used	Old pages had been swapped, which could still add latency even though RAM was available.
CPU idle in top	Approximately 64.7%	The CPU was not continuously saturated.
Load average	Below approximately 1.0	The machine was not under extreme system-wide load.

The measurements suggested that the long wait was not simply a 100% CPU or zero-memory condition. The Citrix web-based interface, Rowan resource synchronization, first-login caching, client compatibility, or a combination of local rendering and server response remained possible causes. Citrix was installed and authenticated, but its Home resource population remained slow.

4.16 Boot regression after the installer USB was reformatted

After the 64 GB Lubuntu USB was reformatted back into ordinary storage, the computer again failed to boot without it. Because the installed Lubuntu system was still running at the time, the disk and bootloader were inspected before rebooting.

lsblk showed the WD Blue as /dev/sda, with /dev/sda1 mounted as /. GRUB was reinstalled from the running installed system and completed successfully:

```
sudo grub-install /dev/sda
sudo update-grub
```

grub-install reported “Installing for i386-pc platform” and “Installation finished. No error reported.” update-grub found the Lubuntu theme, kernels 7.0.0-28-generic and 6.17.0-35-generic, initramfs images, and memtest. Despite this successful result, the next boot still displayed the boot-device error.

Fresh BIOS inspection then revealed that SATA1 Controller Mode had returned to RAID. Changing it back to IDE immediately restored normal booting. This was decisive proof that the successful GRUB installation was not the missing piece.

4.17 Replacing the CMOS battery

Because the SATA setting had returned to RAID more than once, the old CMOS battery was replaced as preventive maintenance. The motherboard battery was found behind the expansion area and was accessible without removing the large tuner/capture card.

Battery	Details
Removed battery	Newsun CR2032, 3 V
Replacement battery	Duracell CR2032, 3 V
Orientation	Positive “+” side facing up; battery pressed flat under the retaining clip.

A plastic spudger was used to push the retaining clip away from the battery. After the new battery was installed, the BIOS displayed a checksum/configuration reset message and loaded factory defaults. RAID returned because RAID was the factory/default SATA mode. SATA1 Controller Mode was therefore set back to IDE and saved again. The computer then returned to the Lubuntu family login screen.

Expected behavior after battery replacement

Removing the CMOS battery clears saved BIOS configuration. A checksum warning and return to factory defaults are normal. The important post-replacement task is to re-enter the BIOS, restore IDE mode, check date/time and boot order, and save.

4.18 Removing and inspecting the original hard drive

The original Seagate hard drive had previously been disconnected from SATA data but remained physically installed. During final internal maintenance, its SATA power and data connections were disconnected, the tool-less cage was released, the cable bundle was moved aside, and the drive slid out. The other internal cables were reconnected afterward.

The drive was then inserted into an Insignia SATA drive dock connected to a Windows 11 Snapdragon laptop. Device Manager recognized it as “ST380013 AS SCSI Disk Device.” The SCSI wording came from the USB-to-SATA bridge presentation and did not mean the physical disk used a SCSI connector.

Windows Disk Management recognized Disk 1 as a 74.53 GB online basic disk containing one Healthy (Active, Primary Partition), but Windows did not assign a drive letter or list a filesystem. CrystalDiskInfo and DiskPart were used to determine why.

5. Final Verified Configuration

Category	Final state
Computer	HP Pavilion a6660t, Intel Core 2 Duo E8400, 3 GB DDR2 RAM.
Primary storage	WD Blue SA510 2.5-inch SSD, approximately 1 TB; Linux device /dev/sda, root partition /dev/sda1.
Original HDD	Seagate ST380013AS 80 GB removed from the case and preserved externally.
Operating system	Lubuntu 24.04 LTS family, updated during the project; legacy i386-pc GRUB installation.
BIOS storage mode	SATA1 Controller Mode = IDE. This is essential.
Boot device priority	Hard Drive Group first; CD-ROM Group second; Floppy Group third; Network Boot Group fourth. HDD Group should identify WD Blue SA510.
CMOS battery	New Duracell CR2032 3 V installed, positive side up.
Wireless networking	TP-Link Archer TX1800U Nano Wi-Fi 6 USB adapter active; old Wi-Fi interface unmanaged.
Wi-Fi interface mapping	Old interface: wlx001644187404. New TP-Link interface: wlxccbabdde9a6a.
Webcam and microphone	EMEET SmartCam C960, Full HD 1080p, fixed focus, built-in microphone.
Speakers	External speakers connected to green 3.5 mm line-out.
Floppy support	External TEAC USB UF000x drive; IBM 1.44 MB disks readable; Mac HFS disks identified.
Productivity software	Firefox, LibreOffice, Cheese, hfsutils, Citrix Workspace. Thunderbird remains unresolved.
Family setup	Multiple family user accounts; only sanjevr verified in sudo group.
Physical deployment	Downstairs living-room family computer.

Capabilities achieved

- Boot Lubuntu from the SSD without the installer USB when BIOS remains in IDE mode.
- Browse the modern web and play YouTube through Firefox.
- Connect to the current home Wi-Fi without changing the router to weaker security.
- Participate in video calls using a 1080p webcam, microphone, and external speakers.
- Open modern and legacy Microsoft Word documents through LibreOffice.
- Read and preserve files from IBM-formatted 3.5-inch floppy disks.
- Identify and investigate Macintosh HFS floppy disks.
- Authenticate to Rowan University’s Citrix Workspace environment.
- Provide separate family accounts while reserving administrator privileges for sanjevr.

6. BIOS Settings and Boot Recovery Runbook

6.1 Essential BIOS settings

BIOS page	Required or expected value	Why it matters
Main	1st Drive: WD Blue SA510 2.5; 2nd Drive: HL-DT-ST DVD-RAM GH15L	Confirms the SSD and optical drive are physically detected.
Advanced	SATA1 Controller Mode: IDE	The decisive setting. RAID produces the boot-device failure on this installation.
Boot Device Priority	1st Hard Drive Group; 2nd CD-ROM Group; 3rd Floppy Group; 4th Network Boot Group	Ensures the BIOS tries the SSD before removable devices.
HDD Group Boot Priority	WD Blue SA510 2.5	Confirms the hard-drive boot group points to the SSD.
Main / Date and time	Current date and time	A reset date can indicate CMOS settings were cleared or lost.

6.2 If “Reboot and Select proper Boot device” returns

1. Do not immediately reinstall Lubuntu or recreate the USB installer.
2. Restart and press F10 at the HP splash screen to enter BIOS Setup.
3. On Main, confirm the WD Blue SA510 is listed as the first drive. If it is missing, power down and inspect the SSD SATA power and data connections.
4. On Advanced, check SATA1 Controller Mode. If it is RAID, change it to IDE.
5. On Boot, confirm Hard Drive Group is first and HDD Group Boot Priority identifies the WD Blue SSD.
6. Press F10, save changes, and exit.
7. Boot with unnecessary storage/floppy USB devices disconnected for the first test; reconnect them after normal boot is confirmed.

Do not misdiagnose a successful GRUB reinstall

grub-install can report success while the computer still fails because the BIOS is presenting the storage controller in RAID mode. Always verify the BIOS before repeating bootloader repairs.

6.3 When the Lubuntu USB is actually needed

The installer USB is useful only if the installed operating system cannot be entered at all and repairs must be performed from a live environment. It is not required for normal operation. If it must be recreated, use Rufus with an MBR/legacy-BIOS-compatible layout.

7. Stress Testing and Verification History

Sanjev repeatedly emphasized that a single successful boot was not enough. The machine was deliberately tested under edge cases so failures would occur during refurbishment rather than later for a family member.

Test	Result in the conversation	What it established
Terminal reboot with sudo reboot	Succeeded after changing RAID to IDE.	Warm reboot path worked.
Graphical Lubuntu Restart	Succeeded after changing RAID to IDE.	Desktop-initiated reboot worked.
Full shutdown and manual power-on	Succeeded after changing RAID to IDE.	Cold boot worked.
Power-loss simulation before battery replacement	Computer booted Lubuntu after being unplugged for about one minute.	The setting survived a short loss of AC power at that time.
Boot with external floppy attached	Earlier failures occurred and BIOS enumeration changed.	Exposed the underlying BIOS/controller problem; USB was not the final root cause.
Boot with installer USB attached	Installed Lubuntu could boot through the USB environment.	Created the misleading boot-anchor theory.
Boot after installer USB was reformatted	Failed until RAID was changed to IDE again.	Proved the USB was masking, not fixing, the BIOS problem.
GRUB reinstall from running Lubuntu	Completed successfully but boot still failed while RAID was active.	Separated bootloader health from BIOS controller configuration.
First boot after CMOS battery replacement	Checksum/config reset; RAID default returned; normal boot after setting IDE.	Confirmed expected reset behavior and the need to reapply IDE.
New Wi-Fi adapter after reboot	Connected to home Wi-Fi.	Confirmed the modern adapter and NetworkManager configuration worked.

Recommended final verification still to perform

- Perform three warm reboots and two cold boots after all hardware is reassembled.
- Unplug AC power for several minutes, reconnect, enter BIOS, and confirm that IDE, boot order, date, and time were retained by the new CR2032.
- Boot once with all normal USB peripherals connected: Wi-Fi adapter, webcam, keyboard, mouse, speakers, and floppy drive.
- Confirm that Wi-Fi reconnects automatically and that the webcam still appears in Cheese or the browser.
- Confirm that Citrix Workspace still launches after the final hardware work.

8. Mistakes, Dead Ends, and Lessons Learned

The troubleshooting record is valuable because it shows how incomplete information can produce plausible but incorrect conclusions. These were not wasted steps: each failure narrowed the diagnosis. The table below separates what was tried, why it seemed reasonable, and what the final evidence showed.

Earlier conclusion or action	Why it seemed plausible	Final correction
Repeatedly repair GRUB	The machine produced a boot-device error and the USB could start the installed system.	GRUB was successfully installed. The BIOS controller remained in RAID mode, so GRUB was not reached normally.
Assume the SSD was not “trusted” by the old BIOS	HDD Group sometimes said Not Installed and the computer behaved inconsistently.	The BIOS could detect the SSD on Main. The incompatibility was the SATA controller mode, not an inability to use a modern SSD.
Use the installer USB permanently as a boot anchor	The machine booted the SSD whenever the installer USB was present.	The USB only provided an alternate boot path. IDE mode removed the need for it.
Try CD-ROM-first boot-order tricks	Older BIOS firmware sometimes falls through device groups differently.	Boot order was secondary. RAID-to-IDE was decisive.
Suggest repartitioning and reinstalling Lubuntu	A classic MBR layout can matter on old BIOS systems.	This destructive step was not needed and was not shown as executed. The existing installation and MBR GRUB were functional.
Assume the floppy USB itself caused permanent boot damage	The failure appeared when the floppy was plugged in.	USB enumeration exposed the issue, but the recurring root setting was RAID.
Confuse the old and new Wi-Fi interface names	Two similar wlx interface names appeared at the same time.	Unplug/replug testing showed wlxccbabdde9a6a was the TP-Link adapter; wlx001644187404 was the old device.
Assume the TP-Link adapter needed a new driver when the first connection dropped	The network disconnected immediately after device changes.	A reboot initialized the adapter cleanly; it then connected without a custom driver build.
Assume apt would install a native Thunderbird deb	apt normally installs repository packages.	On Lubuntu 24.04, thunderbird was a Snap-transition package. PPA pinning would be required for a native deb, and the conversation did not complete that fix.
Use QApt for Citrix dependency resolution	QApt is the graphical package installer for local deb files.	QApt gave only a generic dependency error. apt install ./file.deb resolved and reported dependencies correctly.
Interpret the old HDD as damaged because Windows did not mount it	It had no drive letter and no Windows volume.	DiskPart revealed partition type 83, Hidden Yes, Active Yes: a Linux partition that Windows does not natively mount.

General lessons

- Photograph every BIOS page, not only the page that seems relevant. The Advanced page exposed the decisive setting.
- Separate detection from bootability. The BIOS could detect the SSD while still presenting it in the wrong controller mode.
- A successful command does not prove the diagnosis. grub-install succeeded multiple times, but the machine still could not boot in RAID mode.
- Use controlled tests: warm reboot, cold boot, power loss, and different USB configurations.
- When two devices look similar, identify them by removing one and observing which interface disappears.
- Do not weaken an entire household network to accommodate one old adapter when a local hardware replacement is available.
- Do not format an old disk merely because Windows cannot mount it. Inspect SMART, partition type, and filesystem first.
- A replaced CMOS battery intentionally resets BIOS settings; restore required settings before judging whether the repair worked.

9. Command and Procedure Appendix

9.1 Disk and boot inspection

```
lsblk
lsblk -f
sudo fdisk -l
cat /etc/fstab
sudo grub-install --version
```

In the final running system, lsblk showed /dev/sda as a 931.5 GiB disk and /dev/sda1 mounted at /. Several loop devices were normal Snap package mounts.

9.2 Final successful GRUB refresh from the installed system

```
sudo grub-install /dev/sda
sudo update-grub
```

The successful output included “Installing for i386-pc platform” and “Installation finished. No error reported.” This verified GRUB but did not by itself fix RAID mode.

9.3 Live-USB chroot repair method used earlier

```
sudo mount /dev/sda1 /mnt
sudo mount --bind /dev /mnt/dev
sudo mount --bind /proc /mnt/proc
sudo mount --bind /sys /mnt/sys
sudo chroot /mnt
grub-install /dev/sda
update-grub
exit
sudo umount /mnt/dev
sudo umount /mnt/proc
sudo umount /mnt/sys
sudo umount /mnt
```

This method is appropriate when the installed system cannot boot. In this case, it was not the final solution because BIOS RAID mode remained active.

9.4 Wi-Fi interface inspection and final old-device disablement

```
nmcli device
sudo nmcli radio wifi off
sudo nmcli radio wifi on
sudo nmcli device set wlx001644187404 managed no
```

Final mapping: old adapter wlx001644187404; TP-Link TX1800U Nano wlxccbabdde9a6a.

9.5 Floppy and HFS inspection

```
lsblk
sudo apt install hfsutils -y
sudo umount /dev/sdf
sudo hmount /dev/sdf
hls
```

The USB floppy enumerated as /dev/sdf during the documented HFS test. Device names can change depending on which storage devices are connected, so lsblk should always be run first.

9.6 User and administrator management

```
awk -F: '$3 >= 1000 && $3 < 65534 {print $1}' /etc/passwd
ls /home
getent group sudo
sudo passwd USERNAME
sudo passwd -e USERNAME
sudo deluser USERNAME sudo
```

The verified sudo-group output listed only sanjevr.

9.7 Thunderbird commands attempted

```
sudo apt update
sudo apt install thunderbird -y
sudo snap remove thunderbird
snap saved
sudo add-apt-repository ppa:mozillateam/ppa
sudo apt update
sudo apt install thunderbird
```

The final apt command still selected the Snap transition package. Do not assume Thunderbird is resolved merely because apt reports it installed.

9.8 Citrix installation and launch

```
cd ~/Downloads
sudo apt install ./icaclient*.deb
/opt/Citrix/ICAClient/selfservice
```

Optional App Protection, deviceTRUST, and EPA components were declined.

9.9 Performance checks

```
free -h
top
```

top is interactive. Press q to exit normally or Ctrl+C to interrupt. On this system, terminal rendering occasionally appeared as blank space with blinking blocks until the command was interrupted or rerun.

9.10 Windows DiskPart inspection of the removed HDD

```
diskpart
list disk
select disk 1
list partition
select partition 1
detail partition
detail disk
list volume
```

The old disk was Disk 1 in the documented Windows session. Always confirm disk size before selecting a disk because numbering can change.

10. Original Seagate HDD Analysis

The original Seagate drive was not removed because SMART showed a clear mechanical failure. It was removed because the SSD was faster, larger, quieter, and simpler for the refurbished family computer. The external analysis showed the old disk was in unexpectedly good hardware condition.

10.1 CrystalDiskInfo results

Metric	Observed value
Model	ST380013AS, Seagate Barracuda 7200.7
Reported capacity	80.0 GB
Firmware	3.43
Serial number	4MR1RNYF
Interface through dock	UASP (Serial ATA)
Native transfer standard shown	SATA/150
Buffer size	8192 KB
Health status	Good
Temperature	30 °C
Power-on count	1,625
Power-on hours	1,896 hours, approximately 79 total powered-on days
Reallocated sectors	0
Current pending sectors	0
Uncorrectable sectors	0
UltraDMA CRC errors	0
Spin retry count	0

10.2 Windows Disk Management and DiskPart results

Observation	Meaning
Disk 1, Basic, 74.53 GB, Online	Windows detected the physical disk and its MBR-style partition table.
One 74.53 GB Healthy (Active, Primary Partition)	A partition occupies nearly the entire disk.
No drive letter; “Change Drive Letter and Paths” grayed out	Windows did not recognize a mountable Windows filesystem.
detail disk: Disk ID 406B033C; no volumes	The disk existed and was not read-only, but Windows had no recognized volume object for the partition.
detail partition: Type 83	MBR partition type 0x83, traditionally used for Linux filesystems.
Hidden: Yes; Active: Yes; offset 1,048,576 bytes	The partition was marked hidden and bootable/active, beginning at the conventional 1 MiB alignment.

10.3 Final interpretation of the old HDD

The disk was not missing because it was dead. Windows could read the device and SMART information, but it could not mount the partition because the partition type was Linux type 83. The likely filesystem is ext2, ext3, or ext4. The safest next inspection step is to attach the dock to Lubuntu and mount or examine the partition read-only before formatting or repurposing it.

Do not format yet
The hardware is healthy and the partition may contain old Linux files. Inspect it from Linux and copy anything important before any destructive reuse.

11. Pending Work and Future Maintenance

11.1 Pending or incomplete items

Item	Current status	Recommended next action
Thunderbird	Snap installation did not launch reliably; PPA attempt still selected Snap wrapper.	Either complete apt pinning for the Mozilla Team PPA, use the official Thunderbird binary/Flatpak, or use Gmail in Firefox.
Old Seagate contents	Linux type 83 partition identified; not yet browsed in Lubuntu.	Connect the Insignia dock to Lubuntu, inspect with lsblk -f, and mount read-only before copying files.
Citrix resource loading	Installed and authenticated, but the Workspace Home page loaded very slowly.	Test direct browser launch through access.rowan.edu, compare with another device, and use one published desktop/app to judge actual session performance.
Final post-battery stress test	One successful return to the Lubuntu login screen after restoring IDE.	Repeat warm, cold, and power-loss tests and verify IDE/date/time retention.

11.2 Optional hardware maintenance

- Clean dust from the CPU heatsink, case fan, power-supply vents, and front intake while holding fan blades still during compressed-air use.
- Consider replacing CPU thermal paste only if temperatures or fan noise justify removing the cooler.
- Investigate whether the motherboard supports a practical RAM upgrade beyond 3 GB. Citrix and Firefox would benefit, but compatibility and DDR2 cost should be checked before purchase.
- Keep the removed Seagate in an anti-static bag or safe box labeled with model, serial number, removal date, and the note “Linux type 83 partition - inspect before formatting.”

11.3 Routine software maintenance

- Install normal Lubuntu security and stability updates, then restart when kernels, drivers, or low-level packages are updated.
- Periodically verify that the new TP-Link adapter reconnects automatically and that the old adapter remains unmanaged.
- Back up recovered floppy files and family documents to at least two independent storage locations.
- Do not rely on the old 80 GB HDD as the only copy of any important data, even though SMART currently reports Good.

12. Glossary

Term	Meaning in this project
AHCI	Advanced Host Controller Interface, a modern SATA mode. It was discussed conceptually but not used as the final setting on this BIOS.
BIOS	The pre-operating-system firmware that detects hardware and chooses a boot device. F10 enters this HP's BIOS Setup.
CMOS battery	The CR2032 coin cell that preserves BIOS settings and the real-time clock while AC power is disconnected.
DKMS	A Linux framework that rebuilds third-party kernel modules after kernel changes. The old RTL8812AU driver builds failed through DKMS.
ext2/ext3/ext4	Common Linux filesystems. The removed Seagate has partition type 83 and may contain one of these.
GRUB	The bootloader installed to the SSD. It was refreshed successfully but was not the root cause of the RAID-mode boot failure.
HDD	Hard disk drive with spinning magnetic platters and moving read/write heads. The original Seagate was an HDD.
HFS	Classic Macintosh Hierarchical File System used by the Mac-formatted floppies.
IDE mode	A legacy-compatible way for the BIOS SATA controller to present the SSD. This was the required final setting.
LTS	Long-Term Support. Lubuntu 24.04 LTS receives extended maintenance compared with short-lived releases.
MBR	Master Boot Record, the traditional legacy-BIOS disk boot structure and partition-table style.
RAID mode	Storage-controller mode designed for RAID-capable operation. On this machine, the factory RAID setting prevented the installed system from booting normally.
Rufus	Windows utility used to write the Lubuntu ISO to a bootable USB and select legacy-compatible partition/firmware options.
SATA	Serial ATA, the physical storage interface used by both the Seagate HDD and WD Blue SSD.
SMART	Self-Monitoring, Analysis and Reporting Technology. CrystalDiskInfo used SMART data to assess the old HDD.
Snap	Canonical's sandboxed application packaging system. Ubuntu's Thunderbird package redirected apt installation to a Snap.
SSD	Solid-state drive using flash memory with no moving parts. The WD Blue SSD replaced the slow mechanical HDD.
UASP	USB Attached SCSI Protocol used by the Insignia dock to present the SATA drive efficiently over USB.
UVC	USB Video Class, the standard that allowed the EMEET webcam to work without a special driver.
WPA2/WPA3	Wi-Fi security protocols. The old adapter had compatibility trouble with the home network; the modern adapter solved it without changing router security.

End of conversation-only restoration guide